

Theme 2: Assistive Solutions (EnAble India)

(Engineering for Accessibility, Functional Independence & Universal Inclusion)

SDG Alignment: SDG 3 (Good Health and Well-being), SDG 8 (Decent Work & Economic Growth), SDG 10 (Reduced Inequalities)

ANRF Alignment: Assistive Technologies & Robotics, Bio-Engineering, Human-centric Computing, Cyber-Physical Systems

- **Smart Mobility Aids & Adaptive Transport**
 - Intelligent motorized wheelchairs, stair-climbing mobility aids, smart white canes with obstacle detection, and low-cost adaptive steering attachments for transport vehicles. Eye-tracking communication interfaces, braille-display devices, gesture-to-speech smart gloves, and tactile communication boards for non-verbal or speech-impaired individuals.
- **Augmentative & Alternative Communication (AAC) Hardware**
 - Eye-tracking communication interfaces, braille-display devices, gesture-to-speech smart gloves, and tactile communication boards for non-verbal or speech-impaired individuals.
- **Prosthetics, Orthotics & Physical Assistance Tools**
 - Low-cost myoelectric robotic prosthetics, wearable exoskeletons for limb support, active orthotic braces, and motorized lifting/transfer aids for bed-to-chair mobility.
- **Sensory Augmentation & Assistive Vision/Hearing Tech**
 - Haptic feedback wearables for spatial awareness, smart glasses for sight translation/object recognition, specialized bone-conduction hearing devices, and directional sound amplifiers.
- **Cognitive, Executive & Learning Support Tools**
 - Step-by-step visual, audio, and RFID prompt hubs for independent task execution in Intellectual Disabilities, and multisensory tactile-audio calculation blocks for Dyscalculia and SLD.

- **Sensory Regulation, Autism & Emotional Wellbeing Hardware**
 - Dynamic pneumatic Deep Pressure Therapy (DPT) compression vests, adaptive acoustic environment attenuators, and modular desktop calming/sensory soothing hubs.